

DS	Session	Hashing	Question Information	Level 1	Challenge 100
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Problem Description

Mathison has uncovered an ancient piece of paper on which are written N integers. This set of numbers will be referred to as A . Mathison learned in History class that a group of integers is unique if and only if their total is divisible by a mythological constant M .

Mathison tries to find out how many **distinct** triplets of numbers, from the piece of paper, have their sum divisible by M . Unfortunately, this problem is quite hard to crack and he needs your help.

Constraints

- $1 \leq N \leq 2 \times 10^5$
- $1 \leq M \leq 10^4$
- $0 \leq A[i] \leq 2 \times 10^9$

Input

The first line of the input file contains two space-separated integers, N and M , representing the number of integers and the mythical constant.

The next line contains N space-separated integers, where the i^{th} integer represents $A[i]$.

Output

The output file should contain only one integer, the answer to Mathison's question.

Explanation for Test case 1

There are 26 special trios: {0, 1, 2}; {0, 1, 8}; {0, 2, 5}; {0, 2, 6}; {0, 2, 9}; {0, 3, 7}; {0, 5, 8}; {0, 6, 8}; {0, 8, 9}; {1, 2, 7}; {1, 3, 4}; {1, 5, 6}; {1, 5, 9}; {1, 6, 9}; {1, 7, 8}; {2, 4, 8}; {2, 5, 7}; {2, 6, 7}; {2, 7, 9}; {3, 4, 5}; {3, 4, 6}; {3, 4, 9}; {5, 6, 9}; {5, 7, 8}; {6, 7, 8}; {7, 8, 9};

Note: Here we only show the **positions** (0-indexed) in each trio.

Logical Test Cases

Test Case 1

INPUT (STDIN)

10 5
1 10 4 3 2 5 0 1 9 5

EXPECTED OUTPUT

26

Test Case 2

INPUT (STDIN)

10 5
11 10 14 31 21 15 10 11 9 51

EXPECTED OUTPUT

31

Mandatory Test Cases

Test Case 1

KEYWORD

while(i<k)

✓ Complexity Test Cases

Test Case 1

CYCLOMATIC COMPLEXITY

10

Test Case 2

TOKEN COUNT

450

Test Case 3

NLOC

55